The code shown in the next listing, which fetches the list of the available AMIs, is located at lib/listAMIs.js.

Listing 4.3 Fetching the list of available AMIs: /lib/listAMIs.js

```
const AWS = require('aws-sdk');                          ①
const ec2 = new AWS.EC2({region: 'us-east-1'});          ②

module.exports = (cb) => {                               ③
  ec2.describeImages({                                   ④
    Filters: [{
      Name: 'name',
      Values: ['amzn2-ami-hvm-2.0.202*-x86_64-gp2']
    }]
  }, (err, data) => {
    if (err) {                                           ⑤
      cb(err);
    } else {                                             ⑥
      const amiIds = data.Images.map(image => image.ImageId);
      const descriptions = data.Images.map(image => image.Description);
      cb(null, {amiIds: amiIds, descriptions: descriptions});
    }
  });
};
```

① require is used to load modules.

② Creates an EC2 client

③ module.exports makes this function available to users of the listAMIs module.

④ API call to list AMIs

⑤ In case of failure, err is set.

⑥ Otherwise, data contains all AMIs.

The code is structured in such a way that each action is implemented in the lib folder. The next step to create a virtual machine is to choose which subnet the virtual machine should be started in. You haven't learned about subnets yet, so for now, select one randomly; see figure 4.9.